

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1769

Mud Lake 138 kV

October 26, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1769. The Interconnection Customer proposes a 65.7 MW solar photovoltaic facility interconnecting at the Mud Lake 138 kV substation in the West Virginia area. The request was received on June 06, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 538 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,349,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                            | Contingency                                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------|---------------------------------------------------------------|-------------|--------------|----------|
| Clark County-Shelton 138.0 kV                 | Artesia Country Club-Clark County 345.0 kV                    | 126.2       | 246          | 5.5      |
| Palmyra Substation-Grimmway-Malaga 138.0 kV L | Clark County-Kearneyville 345.0 kV                            | 122.4       | 313          | 9.7      |
| Glenbrook-Palmyra 138.0 kV                    | Kearneyville-Clark County 345.0 kV                            | 121.9       | 325          | 9.7      |
| Redman-Glenbrook Substation 138.0 kV Line     | Clark County-Kearneyville 345.0 kV L                          | 121.5       | 336          | 9.7      |
| Gallegos-Redman 138.0 kV                      | Clark County-Kearneyville 345.0 kV                            | 121.2       | 344          | 9.7      |
| Galegos-Killingsworth 138.0 kV Line           | Clark County-Kerneyville 345.0 kV                             | 120.9       | 354          | 9.7      |
| Montano Substation-Celeron 138.0 kV           | Rio Dell-Celeron 138.0 kV                                     | 119.8       | 377          | 39.0     |
| Shelton-Five Channels 138.0 kV line           | Artesia Country Club-Clark County 345.0 kV                    | 119.6       | 239          | 5.5      |
| Rio Dell-Montano 138.0 kV                     | Rio Dell Substation-Celeron 138.0 kV L                        | 119.6       | 381          | 39.0     |
| Rio Dell-Celeron 138.0 kV L                   | Rio Dell Substation-Montano 138.0 kV                          | 119.6       | 361          | 37.0     |
| Air Force Base 345/138kV Transformer          | Tiger Bay-Air Force Base 345.0 kV Line                        | 113.2       | 541          | 6.5      |
| St. John-Mud Lake 138.0 kV                    | Prospect Central-Marvin Griffin Road 138.0 kV L               | 110.8       | 242          | 48.3     |
| Garden Corners-Arvin 138.0 kV                 | Southwest Research Institute-Douglas City Substation 138.0 kV | 108.7       | 171          | 5.6      |
| St. John-Rio Dell 138.0 kV                    | International Paper Valliant Ok-Kearneyville 345.0 kV         | 107.0       | 593          | 40.6     |
| Douglas City-Garden Corners 138.0 kV Line     | Southwest Research Institute-Douglas City 138.0 kV L          | 106.8       | 214          | 5.6      |

| Monitored Facility                                 | Contingency                                           | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------|-------------------------------------------------------|-------------|--------------|----------|
| Shuffleton-Arvin 138.0 kV                          | Cark County-Karneyville Substation 345.0 kV L         | 106.2       | 223          | 7.5      |
| Grimmway-Malaga-Sleepy Hollow 138.0 kV             | Clark County-Kearneyville 345.0 kV line               | 105.5       | 487          | 8.9      |
| Millville-Sleepy Hollow 138.0 kV                   | Clark County-Kearneyville 345.0 kV                    | 105.3       | 474          | 8.9      |
| Douglas City-Millville 138.0 kV L                  | Clark County-Karneyville 345.0 kV                     | 105.3       | 461          | 8.9      |
| Southwest Research Institute-Douglas City 138.0 kV | Kearneyville-Clark County 345.0 kV                    | 101.9       | 261          | 5.4      |
| Arvin-Southwest Research Institute 138.0 kV        | Clark County-Kearneyville 345.0 kV                    | 101.8       | 226          | 5.4      |
| N Mesa Substation-Sandalfot Substation 138.0 kV    | Kearneyville-International Paper Valliant Ok 345.0 kV | 101.4       | 425          | 18.1     |
| Celeron Substation-Sandalfot 138.0 kV              | International Paper Valliant Ok-Kearneyville 345.0 kV | 100.9       | 442          | 18.1     |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                      | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Clark County Substation-Shelton 138.0 kV      | Rebuild line with higher-capacity conductor | \$2,053,000   | 7% (\$145,000)        |
| Palmyra Substation-Grimmway-Malaga 138.0 kV L | Rebuild line with higher-capacity conductor | \$2,556,000   | 18% (\$449,000)       |
| Glenbrook-Palmyra 138.0 kV                    | Rebuild line with higher-capacity conductor | \$2,027,000   | 18% (\$356,000)       |
| Redman-Glenbrook 138.0 kV                     | Rebuild line with higher-capacity conductor | \$2,611,000   | 18% (\$459,000)       |
| Gallegos-Redman 138.0 kV line                 | Rebuild line with higher-capacity conductor | \$1,931,000   | 18% (\$339,000)       |
| Gallegos-Killingsworth 138.0 kV               | Rebuild line with higher-capacity conductor | \$6,039,000   | 18% (\$1,062,000)     |
| Mntano Substation-Celeron 138.0 kV            | Rebuild line with higher-capacity conductor | \$1,245,000   | 68% (\$851,000)       |
| Shelton-Five Channels 138.0 kV line           | Rebuild line with higher-capacity conductor | \$1,959,000   | 7% (\$138,000)        |
| Rio Dell-Montano 138.0 kV                     | Rebuild line with higher-capacity conductor | \$2,115,000   | 68% (\$1,446,000)     |
| Rio Dell-Celeron 138.0 kV L                   | Rebuild line with higher-capacity conductor | \$3,989,000   | 100% (\$3,989,000)    |
| Air Force Base 345/138kV Transformer          | Replace transformer with higher-rated unit  | \$11,868,000  | 7% (\$779,000)        |

| Facility                                           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| St. John-Mud Lake 138.0 kV                         | Rebuild line with higher-capacity conductor | \$10,196,000  | 100% (\$10,196,000)   |
| Arvin-Garden Corners 138.0 kV                      | Reconductor line                            | \$3,414,000   | 31% (\$1,062,000)     |
| St. John-Rio Dell 138.0 kV                         | Reconductor line                            | \$2,453,000   | 63% (\$1,556,000)     |
| Douglas City-Garden Corners 138.0 kV Line          | Reconductor line                            | \$763,000     | 31% (\$237,000)       |
| Arvin-Shuffleton 138.0 kV L                        | Reconductor line                            | \$905,000     | 13% (\$114,000)       |
| Grimmway-Malaga-Sleepy Hollow 138.0 kV             | Reconductor line                            | \$874,000     | 13% (\$110,000)       |
| Millville-Sleepy Hollow 138.0 kV                   | Reconductor line                            | \$2,184,000   | 13% (\$275,000)       |
| Douglas City-Millville Substation 138.0 kV L       | Reconductor line                            | \$4,819,000   | 13% (\$607,000)       |
| Southwest Research Institute-Douglas City 138.0 kV | Reconductor line                            | \$740,000     | 100% (\$740,000)      |
| Arvin-Southwest Research Institute 138.0 kV        | Reconductor line                            | \$3,034,000   | 100% (\$3,034,000)    |
| Sandalfoot-N Mesa 138.0 kV                         | Reconductor line                            | \$945,000     | 22% (\$206,000)       |
| Celeron-Sandalfoot 138.0 kV                        | Reconductor line                            | \$911,000     | 22% (\$199,000)       |
|                                                    | Total allocated to project                  |               | \$28,349,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.